

Zhuoyan Sun

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Education

University College London

Sep 2025 - Present

MSc. Health Data Science (1st Degree)

London | UK

- **Core Modules:** Data Science and Statistics, Artificial Intelligence in Healthcare, Advanced Machine Learning for Healthcare, Advanced Data Science and Statistics, Artificial Intelligence in Healthcare Group Project

Imperial College London

Sep 2021 - Jun 2024

BSc. Biological Sciences

London | UK

- **Core Modules:** Mathematics, Bioinformatics, Programming and Statistics, Computational Omics, Integrative System Biology
- **Final Year Project:** Structural and Dynamic Analysis of Rare Genetic Variants in C9 and their Membrane-Binding Mechanisms Revealed by Molecular Dynamics Simulations

Work Experience

Data Scientist (AI agent)

Jul 2026 - Present

DS Digital Technology

Hangzhou | CN

- Developed AI Agent and RAG prototypes for enterprise knowledge retrieval, workflow automation, and decision support
- Implemented document processing, embedding retrieval, prompt design, LLM generation, evaluation, and API integration
- Supported client-facing AI solution design through data analysis, metric definition, and cross-functional collaboration

Associate Strategy Analyst

May 2025 - Aug 2025

Ipsos Strategy3

Beijing | CN

- Analyzed large-scale healthcare and consumer datasets from 100+ expert interviews to extract quantitative trends and evidence-based market insights
- Performed data-driven market sizing and predictive modeling to estimate \$20M revenue potential across emerging health product segments
- Collaborated with cross-functional data, research, and strategy teams to translate analytical findings into business recommendations and executive dashboards

Project Experience

Clinical NLP & LLM Pipeline for Radiology Report

Jan 2026 - Present

University College London

London | UK

- Built a NLP pipeline using Hugging Face Transformers to parse radiology reports and generate structured features for downstream ML tasks
- Fine-tuned ClinicalBERT-style models for representation learning on clinical text, enabling quantification of diagnostic difficulty and semantic understanding
- Developed a model evaluation framework to surface failure modes in vision-language systems and benchmark reliability using text-derived signals and patient data

Causal ML Model for Cardiovascular Risk Simulation

Dec 2025 - Present

University College London

London | UK

- Built an end-to-end ML pipeline on longitudinal biobank data, engineering multi-timepoint features to predict cardiovascular risk and model behavioural dynamics
- Developed and validated both predictive and causal ML models to generate individual-level counterfactual risk estimates for lifestyle interventions
- Designed a simulation-driven webpage translating model outputs into actionable, user-facing insights for preventive healthcare

Skills

Programming Skills: Python, JavaScript, R, SQL, MySQL, Linux Shell

Data&Visualization: pandas, NumPy, Matplotlib, Seaborn, Power BI, Excel

ML&AI: Scikit-learn, PyTorch, TensorFlow, Hugging Face Transformers, XGBoost

MLOps&Tools: Docker, Git, GitHub, Linux, Streamlit, AWS

Language: English, Mandarin